Technology-Assisted Data Annotation for Objective and Subjective Tasks

Doctoral Consortium Paper

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Abstract

Large-scale annotated datasets are a fundamental component of modern Natural Language Processing (NLP) systems, yet manual annotation remains expensive and time-consuming. Recent advances in Machine Learning (ML) and Artificial Intelligence (AI) enabled technology-assisted annotation pipelines that combine automatic labeling with human supervision. However, the design and evaluation of such systems depend strongly on the nature of the annotation task. My doctoral research studies technology-assisted annotation for textual data, focusing on Objective Annotation Tasks (OATs), where labels correspond to factual or schema-driven properties, and Subjective Annotation Tasks (SATs), where labels depend on interpretation and individual judgment. For OATs, I investigate Human-In-The-Loop (HITL) pre-annotation workflows and fully automatic annotation pipelines with statistically grounded sampling-based validation. For SATs, my work focuses on analyzing, decomposing, and understanding variability in multi-annotator datasets. The final goal of this research is to develop a task-aware annotation framework supporting different annotation support, quality-control, and validation strategies, improving efficiency while remaining reliable and tailored to the intended use of the dataset.

Keywords

Information Extraction (IE), Natural Language Processing (NLP), Data Annotation, Technology-assisted Data Annotation, Automatic Data Annotation, Objective Annotation Tasks (OATs), Subjective Annotation Tasks (SATs)

1. Introduction

Large-scale annotated corpora are essential for building and evaluating Natural Language Processing (NLP) systems. The development of automatic systems for NLP and Information Extraction (IE) over textual data–including token classification tasks such as Named Entity Recognition (NER) and Part-Of-Speech (POS) tagging, and sequence classification tasks such as sentiment analysis and spam detection–relies heavily on high-quality labeled data. However, manual annotation is often expensive, time-consuming, cognitively demanding, and in many domains requires domain-specific expertise [1, 2]. Recent progress in NLP introduced new opportunities to support human annotators through technology-assisted annotation pipelines based on Human-In-The-Loop (HITL), active learning, and automatic annotation systems [3, 4, 5, 6]. These approaches can reduce labeling effort while increasing annotation throughput and consistency. However, they also introduce methodological questions concerning reliability, bias, transparency, and cost-quality trade-offs [7, 8].

My doctoral research investigates how technology-assisted annotation systems for textual data should be designed and evaluated across two different annotation settings. The first concerns Objective-Annotation Tasks (OATs), especially in biomedical and clinical text annotation, where domain-specific terminology requires expert annotators, making large-scale annotation particularly costly. In this context, I study how to effectively implement HITL pre-annotation workflows and fully automatic annotation pipelines combined with statistically grounded sampling-based validation. The second concerns Subjective-Annotation Tasks (SATs), including Hate Speech (HS), toxicity, and offensiveness annotation. In this setting, the focus shifts to understanding disagreement, quantifying annotator

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variability, and evaluating whether diversity-aware automatic annotation support can meaningfully reflect patterns observed in human judgments.

The broader goal of this research is to develop reliable, efficient, and fair technology-assisted annotation frameworks that adapt to the characteristics of the annotation task, combining efficiency-oriented automation with statistically grounded methods for validation and analysis.

2. Background

My research builds on the distinction between OATs and SATs (cf. Table 1), while treating them as idealized endpoints of a continuum along which annotation tasks can be positioned. OATs involve labels that correspond primarily to factual, structural, or schema-driven properties of the text. In this setting, label variation is typically treated as *disagreement*, denoting mutually incompatible annotations that cannot all be correct under the adopted schema [9]. Such disagreement may arise from errors, insufficient expertise, unclear guidelines, or attention lapses. Annotation support aims to reduce curation effort and disagreement without compromising quality or introducing anchoring biases [10]. On the other hand, labels in SATs depend more on individual interpretation, social norms, and personal experiences. Therefore, differences in SAT annotations are referred to as *variability*, and can arise from item ambiguity, annotator uncertainty, different interpretations, or the existence of multiple acceptable labels [9]. However, even OATs may exhibit persistent variability on difficult or underspecified cases [11]. At the same time, even highly subjective tasks may contain clear-cut items on which most annotators agree. The treatment of subjectivity also depends on the annotation protocol: prescriptive approaches seek to minimize individual variation and encode a shared interpretation, whereas descriptive approaches preserve individual judgments to represent different beliefs and perceptions [12].Consequently, annotation support should be evaluated not only in terms of efficiency, but also according to whether it appropriately reduces error while preserving the forms of variability relevant to the intended use of the dataset [13, 14, 15].

Across both tasks, the literature on technology-assisted annotation proposes several paradigms. HITL workflows typically rely on pre-annotation systems, in which models propose initial labels that humans revise, often combined with active-learning strategies that iteratively update models as new labels are curated [16, 17, 18]. Other approaches focus on selective annotation, prioritizing examples for manual curation based on estimated difficulty [5, 19]. More recent work explores systems combining labels with natural-language explanations and agent-based architectures where Large Language Models (LLMs) iteratively reason, justify, and revise predictions in HITL workflows [8, 20, 21].

Concerning SATs, the literature is strongly focused on identifying and disentangling the different sources of annotation variability. Such variability may reflect meaningful differences in individual judgments, annotator-specific tendencies, or differences in the interpretation of the task and of linguistic phenomena such as irony and sarcasm. However, as in OATs, it may also arise from annotation errors, insufficient attention, or misunderstandings of the guidelines. Decomposing and classifying these sources is therefore essential for distinguishing meaningful variability from noise, improving dataset curation and post-processing, and determining how individual judgments should be treated under prescriptive or descriptive annotation paradigms [9, 12, 22, 23].

3. Research Directions

Building on the discussion presented in Section 2, my doctoral research is based on the following Research Questions (RQs) (see Figure 1):

RQ1. How can HITL pre-annotation workflows reduce the cost of OATs while preserving reliability?

RQ2. How should automatic annotation systems be introduced and reliably validated?

RQ3. How should technology-assisted annotation systems be designed to appropriately support SATs?

Table 1
Typical characteristics of objective and subjective annotation tasks

Aspect	Objective Annotation Tasks (OATs)	Subjective Annotation Tasks (SATs)
Protocol	Usually prescriptive	Prescriptive or descriptive, depending on the intended use
Labels	Factual or schema-driven	Interpretation- and perception-dependent
Expected Label Variation	Labels largely deterministic	Labels possibly varying across annotators
Sources of Variation	Text difficulty, unclear guidelines, unclear guidelines, annotation error	Item ambiguity, individual interpretation and tendencies, unclear guidelines, annotation error
Treatment of Variation	Investigated to identify errors or unresolved ambiguity	Decomposed to distinguish meaningful variability from disagreement and noise
Goal of Annotation Support	Improve annotation efficiency while preserving quality and correctness	Reduce error while either constraining or preserving variability
Typical Support Paradigm	HITL pre-annotation, iterative/active learning, automatic annotation with validation	Variability-aware analysis and support
Evaluation Focus	Accuracy, consistency, cost-quality trade-offs	Variability structure, noise separation
Support-related Risks	Propagation of model errors, anchoring biases	Suppression of meaningful variations, anchoring biases
Typical tasks	NER, NEL, RE	Hate speech, toxicity, offensiveness, sexism

For OAT, my research studies two levels of technology support. The first (**RQ1**, blue rectangle in Fig. 1) focuses on HITL pre-annotation, where automatically generated labels are revised and validated by human annotators. Effectiveness is evaluated in terms of annotation quality, measured with standard NLP metrics (Precision, Recall, F1-score) and Inter-Annotator Agreement (IAA) measures (e.g., Cohen's κ, Fleiss' κ, Krippendorff's α) [24, 25, 26]. Annotation effort is assessed through controlled comparisons of batches annotated with and without pre-annotations or with pre-annotations of varying quality. This also enables the assessment of potential anchoring effects introduced by model suggestions [27]. The second level (**RQ2**, green rectangle in Fig.1) is fully automatic annotations with efficient sampling-based quality estimation. In this setting, labels are automatically produced for a large collection, and only a relatively small subset is manually validated, enabling efficient estimation of overall annotation quality with statistical guarantees. This requires combining statistical sampling methods with designs that ease manual validation. An additional expansion is to evolving collections, where the set of annotations grows incrementally over time. In such cases, previously estimated quality is used as prior information to estimate the quality of the updated collection while minimizing additional review effort [28]. These two support levels can be integrated into an end-to-end pipeline for technology-assisted annotation in OATs, as shown in Fig. 2.

For SATs (**RQ3**, red rectangle in Fig. 1), the main objective is to analyze and decompose observed

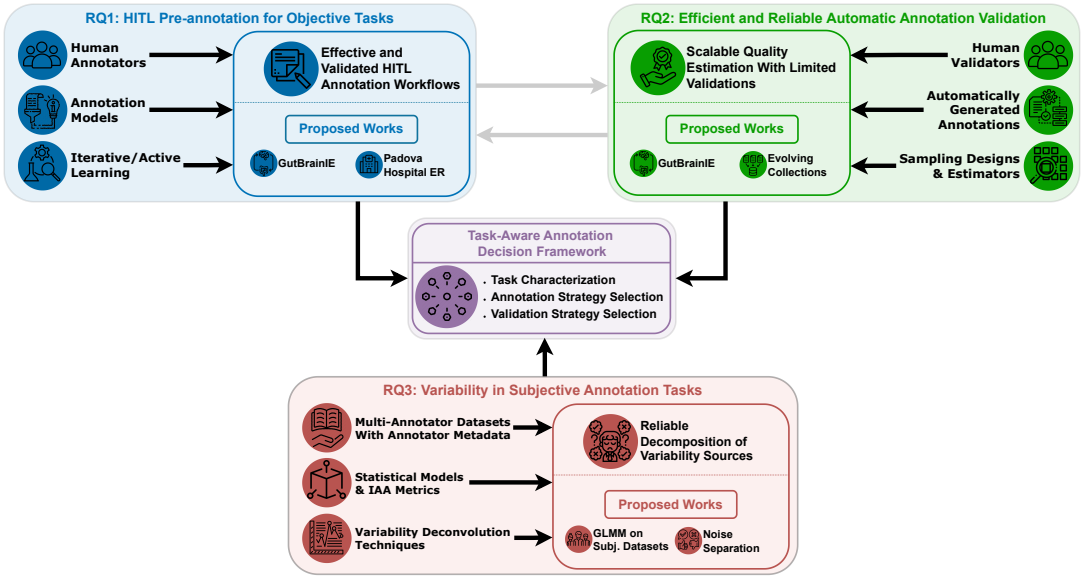


Figure 1: Research project framework based on the defined RQs. The blocks illustrate the main methodological components and research directions, while "Proposed Works" represents the case studies conducted within each research axis.

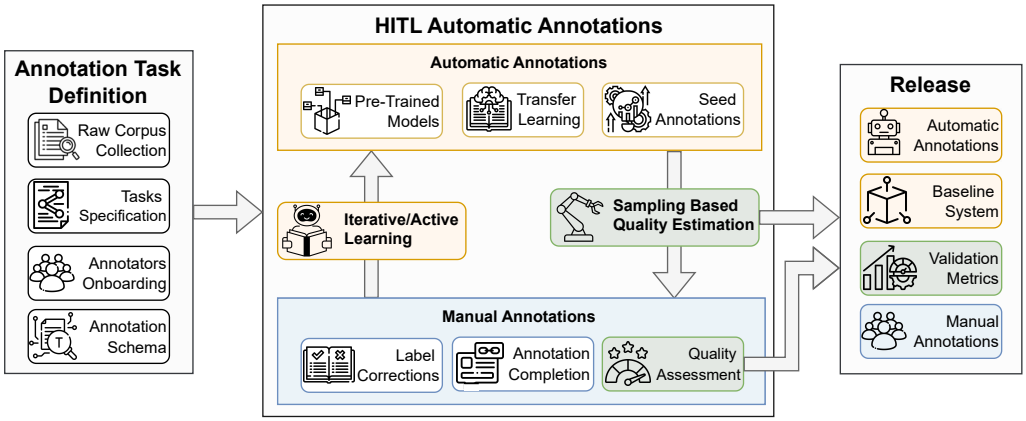


Figure 2: Workflow for OATs integrating HITL pre-annotations with automatic annotations. Colors link components in the central block to the corresponding outputs in the release block.

annotation variability. As shown in Fig. 3, the framework considers datasets with multiple independent annotations per instance, together with annotator-level metadata [29, 30, 31, 32]. The analysis aims to distinguish meaningful and reproducible variation in judgment from disagreement caused by annotation errors, inconsistent behavior, or misunderstandings of the task and guidelines. Standard IAA measures provide an overall characterization of label differences, but cannot identify their underlying sources. They are therefore complemented by disagreement-deconvolution techniques and Generalized Linear Mixed Models (GLMMs), which enable the separation of noisy from meaningful variability and its decomposition into item-level effects, individual annotator tendencies, and mesoscopic factors such as demographic attributes or expertise [22, 13, 15]. The resulting variability profile can guide interventions including guideline revision, annotator training, selective adjudication, recruitment, and the preservation of multiple judgments.

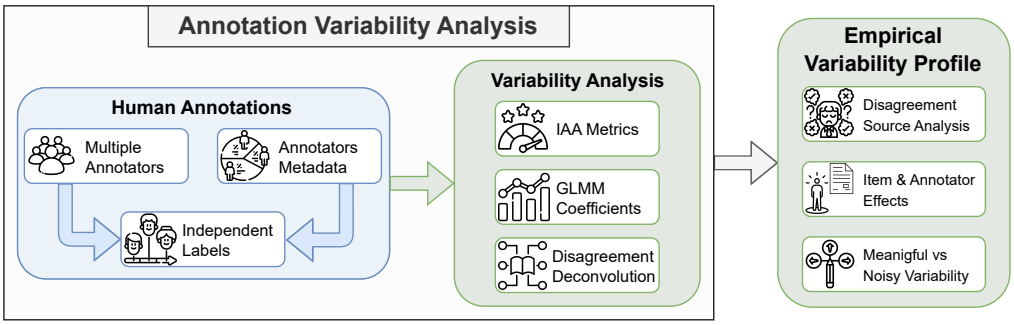


Figure 3: Proposed framework for analyzing and decomposing annotation variability in SATs using multi-annotator datasets.

3.1. Preliminary Work

A first contribution of my work is on OAT, specifically in biomedical IE with the development of the GutBrainIE benchmark, an NLP and IE resource focused on scientific literature about the gut-brain axis, an emerging research field investigating the connections between the gut microbiota, mental health, and neurodegenerative diseases [33, 10]. In this domain, annotation is expensive and cognitively demanding, as it involves fine-grained biomedical concepts and substantial lexical complexity, requiring field expert involvement throughout the annotation and validation process. In this context, my work has contributed to the development of HITL pre-annotation pipelines (RQ1), demonstrating improvements in annotation efficiency and workload reduction, with pre-annotations reducing annotation time by approximately 50% compared to fully manual annotation with high IAA for NER (Fleiss's κ = 0.89, Cohen's mean pairwise κ = 0.88). In parallel, I studied a complementary scenario in which annotations

are automatically generated for the entire collection, and quality is assessed through statistically grounded sampling-based evaluation methods (RQ2) [34]. In this work, I adapt approaches originally developed for efficient accuracy estimation in Knowledge Graphs (KGs) to NER and Named Entity Linking (NEL), proposing a general framework and sampling designs that both reduce the number of manual checks and structure the validation process to lower cognitive load and speed up verification [28, 35]. This framework has been validated on automatically generated NEL annotations for the GutBrainIE collection, showing that overall accuracy can be reliably estimated by manually verifying less than 25% of the annotations. Moreover, reviewing annotations under a tailored sampling design, instead of sequentially document by document, reduced verification time by approximately 30%.

A second OAT case study is an ongoing collaboration with the Emergency Medicine Department of the University Hospital of Padova, focusing on admission and discharge letters. These texts are challenging due to shorthand notation, irregular writing style, abbreviations, acronyms, and context-dependent clinical language. In this project, we are developing an iterative pipeline inspired by GutBrainIE: first defining the annotation schema, then implementing automatic annotation components from a small expert-annotated seed set, and finally refining the system through human correction (RQ1). An initial component for implicit section identification has already been developed and validated by clinicians. Current work focuses on iterative pre-annotation for NER and, subsequently, automatic NEL combined with statistically grounded validation (RQ2).

A third direction of preliminary work concerns SATs, particularly online content-moderation tasks such as HS, toxicity, offensiveness, and sexism. In this setting, I analyze multi-annotator datasets with demographic metadata using GLMMs (RQ3). Across multiple datasets, the emerging pattern is that variability is primarily driven by the item itself, followed by individual annotator tendencies, while demographic variables often contribute little or no explanatory power. Consistent with prior work on SATs, this finding challenges the assumption that attributes such as age, gender, or political orientation are sufficient to explain meaningful annotator diversity [36, 15]. Building on these results, I will apply disagreement-deconvolution techniques to distinguish meaningful variability from annotation noise and assess whether the same variance structure persists once unreliable or inconsistent variation is isolated. This analysis will help determine whether current dataset-construction practices adequately capture meaningful differences in judgment or instead conflate them with artifacts of the annotation process.

4. Conclusion

This doctoral research investigates technology-assisted annotation for textual data across annotation settings with different degrees and sources of label variability. Its main claim is that annotation technologies should not be judged only by how much workload they replace, but by how appropriately they align with the annotation protocol and the intended use of the resulting dataset. In this context, I expect my doctoral research to contribute at two levels: (i) **methodologically**, by developing a task-aware framework that treats OATs and SATs as endpoints of a continuum and uses the empirical structure of annotation variability to guide the selection of support, quality-control, and validation strategies; and (ii) **practically**, by developing and evaluating HITL and automatic annotation workflows for expert domains, particularly biomedical and clinical annotation, where pre-annotation and statistically grounded validation can make large-scale curation more feasible.

These contributions aim to support annotation workflows that adapt to the observed characteristics of each annotation setting, improving efficiency without imposing an inappropriate notion of correctness or preserving uninformative disagreement.

Future work will extend the statistical framework for efficient accuracy estimation to evolving collections, where annotations are produced and updated incrementally. A second direction will investigate its adaptation to SATs to develop strategies that reduce annotation costs or allocate the available budget more effectively across items and annotators while preserving meaningful variability in human judgments.

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Declaration on Generative AI

During the preparation of this work, the author used GPT-5.2 and Grammarly in order to: Grammar and spelling check. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the publication's content.

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